GROUP 5 — MILESTONE 2 (29 OCT 2026)

Milestone 2: Dataset Preparation & Audit

181,231 Photos Across 289 Indian Species & SAM 2.1 Masks

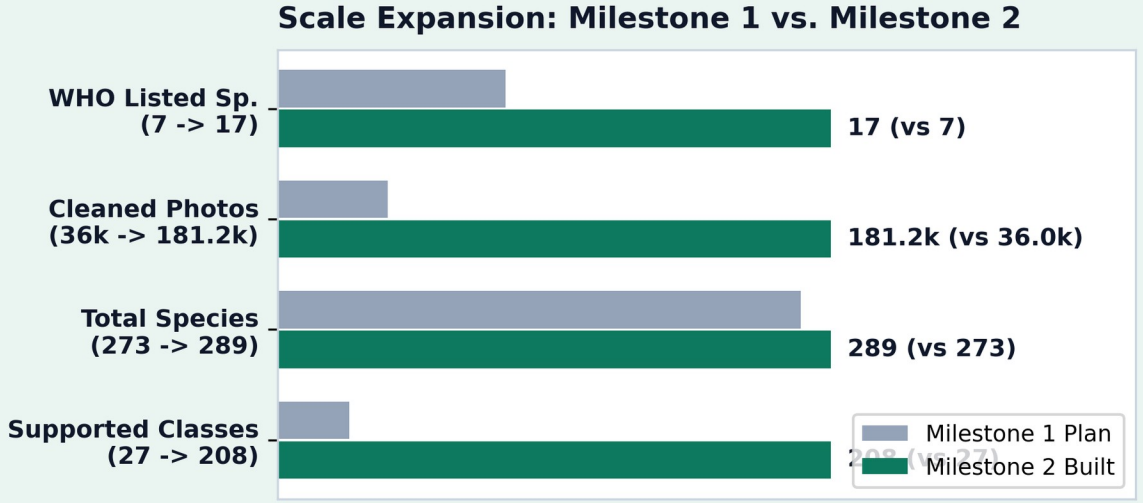
Team Lead: Jayesh Bansal | Members: Vikas, Puneet Bajaj, Manas Rastogi

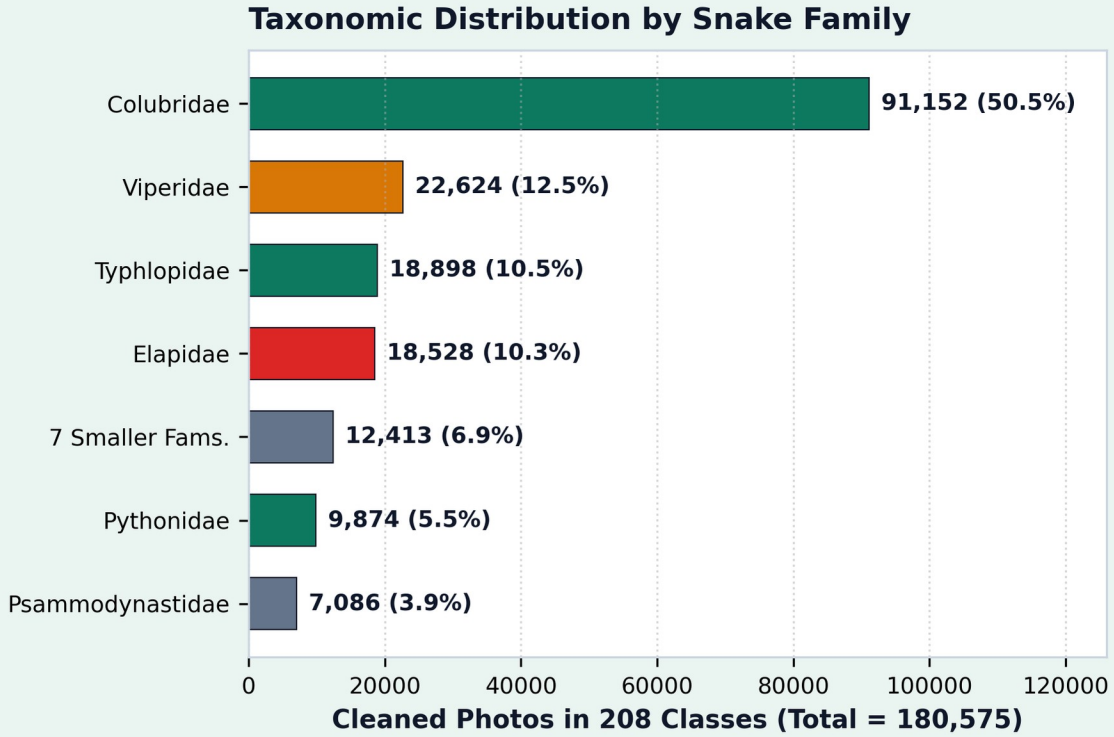
1. DATASET SUMMARY & EXPANSION FROM MILESTONE 1

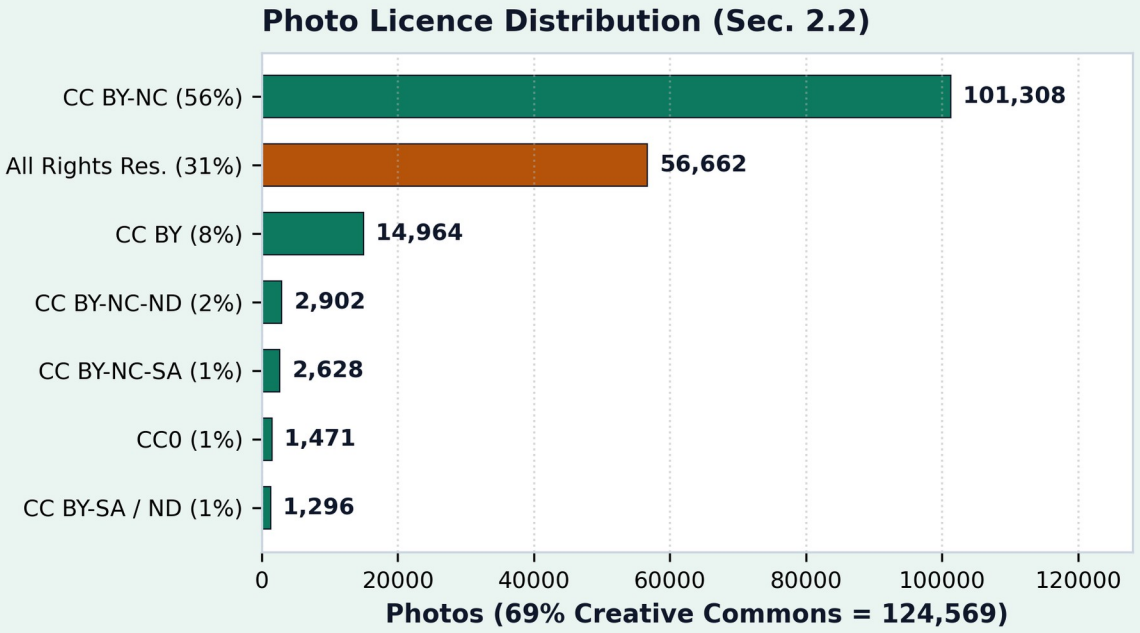
181,231 Cleaned Photos Across 289 Species

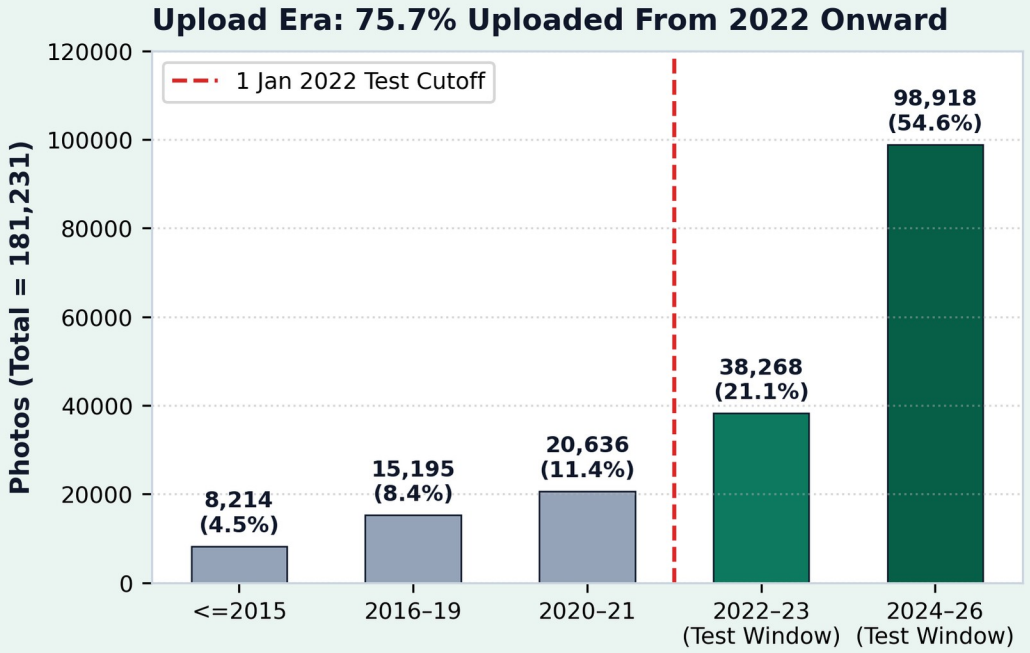
Responding to TA Scope Feedback

- **Expanded Beyond 27 Classes:** Widened from 27 M1 classes (m1_core27) to all 289 Indian species (95,787 obs., 25,041 users).
- **208 Active Classes + Unknown:** 180,575 photos in 208 classes (68 Tier-1, 140 Tier-2) + 656 photos across 81 'unknown' species.
- **Worldwide Train, Indian Test:** Includes global photos of rare Indian snakes (King Cobra, Banded Krait); test stays 100% Indian.









2. SOURCE AUDIT, TAXON-ID QUERIES & IMAGE SPECS

Collection, Licences & Attributes

182,497 Raw Photos in 150 Minutes

- **Zero-Failure Download:** Queried by taxon ID (place 6681, taxon 85553); 24.5 GB at <=500px; 30/30 SHA-1 verified.
- **69% Creative Commons:** 124,569 CC photos (56% CC BY-NC, 8% CC BY) + 31% ARR in private academic Kaggle storage.
- **Key Image Attributes:** Median 500×375 px (125 KB); 1.89 photos/obs; 37.7% taken in India; 18.2% obscured GPS.

3. DEDUPLICATION, QUALITY FILTERING & EXTERNAL SOURCE AUDIT

Cleaning Pipeline: From 182,497 Raw to 181,231 Final Photos

Pipeline Stage / Source	Photo Count	Rule / Filtering Applied	Why It Matters for Model Integrity
1. Raw iNaturalist Download	182,497	289 Indian species, 96,187 observations	Complete worldwide coverage of Indian taxa
2. Exact & Conflict Dupes	-958	928 byte-identical SHA-1 + 30 label conflicts	Prefers Indian copy; drops disputed species
3. Tiny & Cross-Split pHash	-308	36 tiny (<5KB/<100px) + 272 split pHash matches	Prevents resized copies inflating test scores
4. Final Processed Dataset	**181,231**	**180,575 in 208 classes + 656 in unknown_pool**	**0 SHA-1 mismatches; 100% verified decode**
5. External Sources Audit	Deferred / Excluded	IBP (HTTP 403), SnakeCLEF 21 (66GB), Kaggle	Avoids unverified labels & test set leakage

4. HANDLING LONG-TAIL DISTRIBUTION & MEDICAL RISK (SECTION 3.2)

Three-Tier Class Structure & 17 WHO Medically Important Species

1. Tier 1 Core

- **68 Classes (>=400):** 162,770 photos (max: I. braminus 18,340).
- **Full Species ID:** Covers 11 of 17 WHO Cat. 1 & 2 species.

2. Tier 2 Rare

- **140 Classes (20-399):** 17,805 photos (mean 127 photos/class).
- **Few-Shot Fallback:** Covers 6 rare WHO species (e.g. O. kaalinga).

3. Unknown Pool

- **81 Species (<20):** 656 photos kept out of all training.
- **Abstention Test:** Evaluates 'unknown' rejection rate.

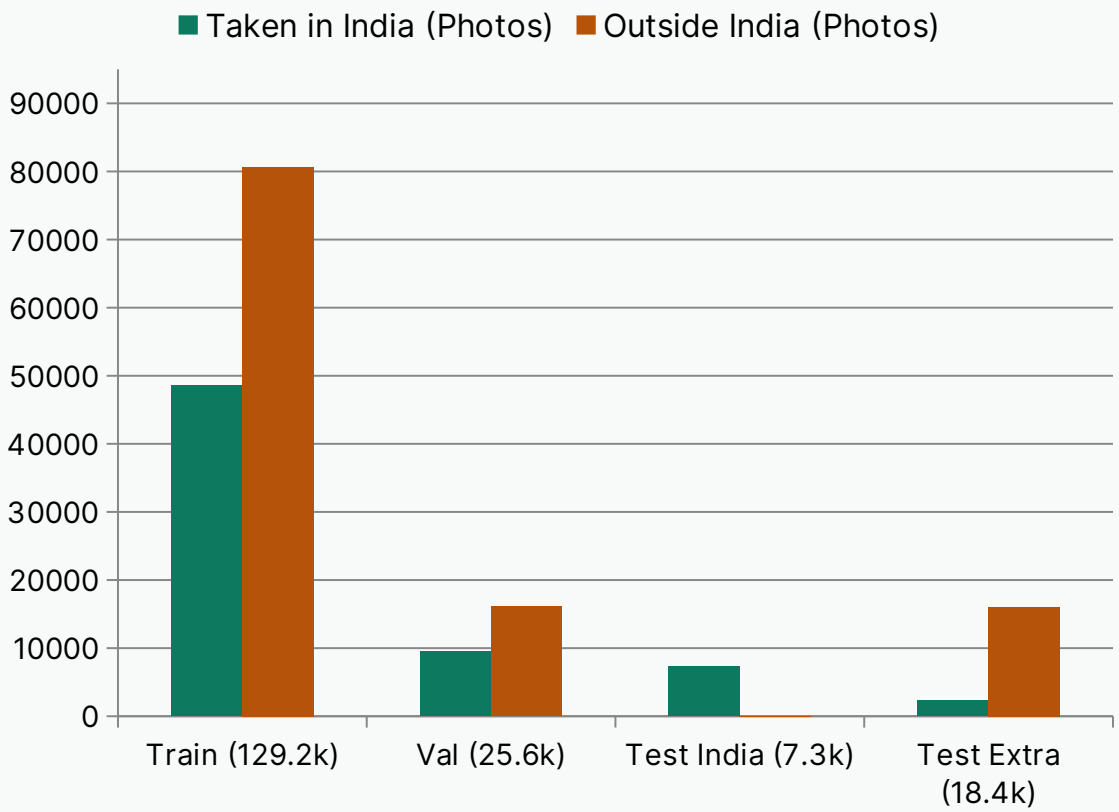
Slide 5 | WHO Category Breakdown (180,575 Photos): Cat. 1 = 13,630 (7.5%) | Cat. 2 = 9,332 (5.2%) | Not Listed = 157,613 (87.3%)

5. STRATIFIEDGROUPKFOLD (7 FOLDS, SEED 42, SCIKIT-LEARN 1.9.1)

Photographer-Grouped Splits & Zero-Leakage Verification

Zero Overlap Across All 4 Checks

- **Photographer Grouping:** 17,856 train, 3,583 val, 767 test, and 2,921 test_extra users with 0 user or obs. overlap.
- **Strict Indian Test Set:** 7,305 photos taken in India and uploaded >=1 Jan 2022 (median 164 photos across M1 core 27).
- **Why classes.csv Matters:** 28 classes have 0 photos in test/; custom loader maps fixed class_id instead of folder index.

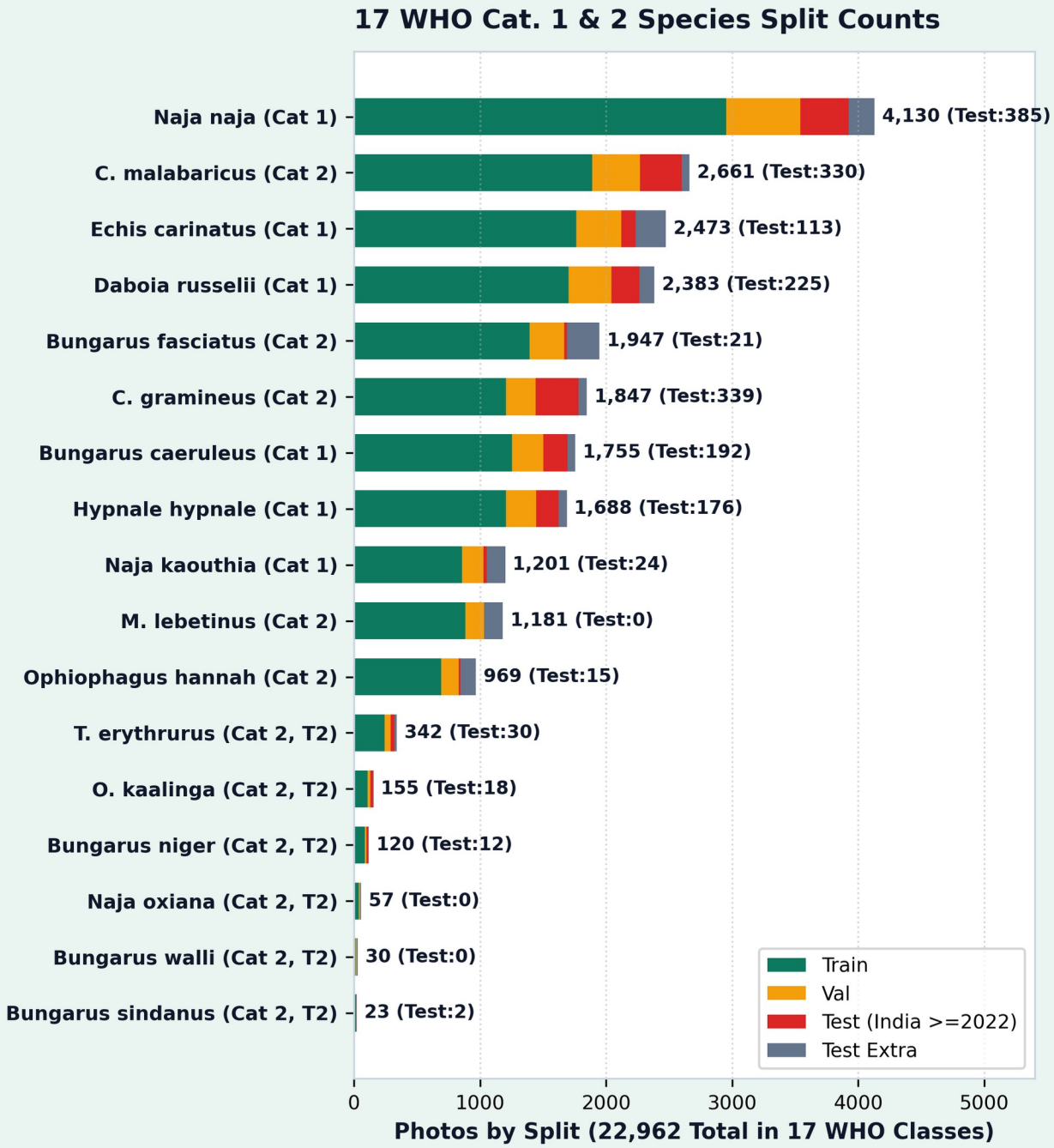


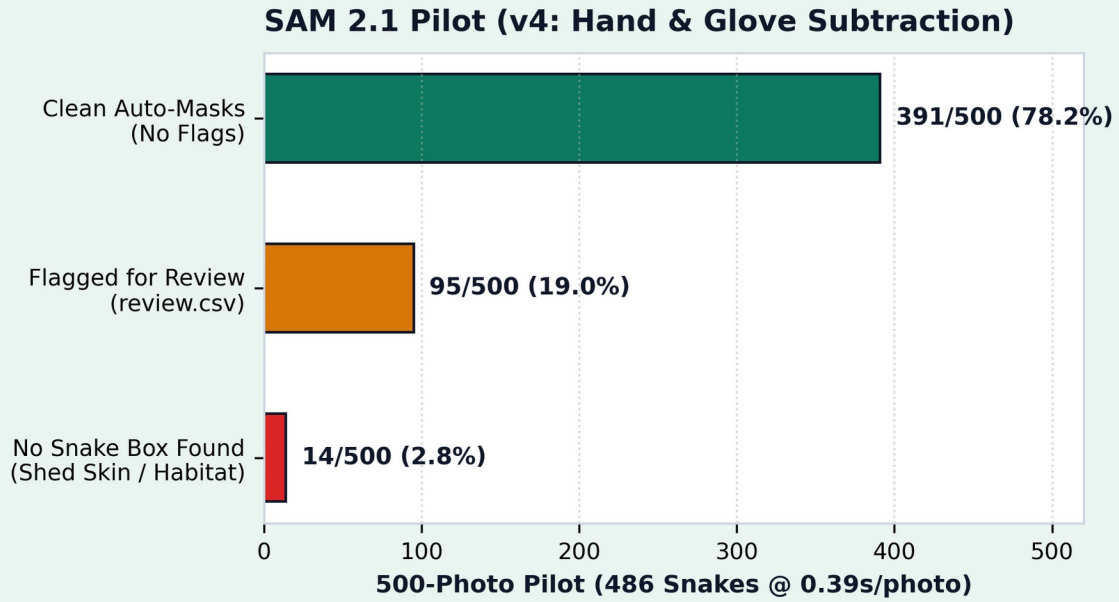
6. SEGMENTATION SUPERVISION & 500-PHOTO PILOT (SECTION 9)

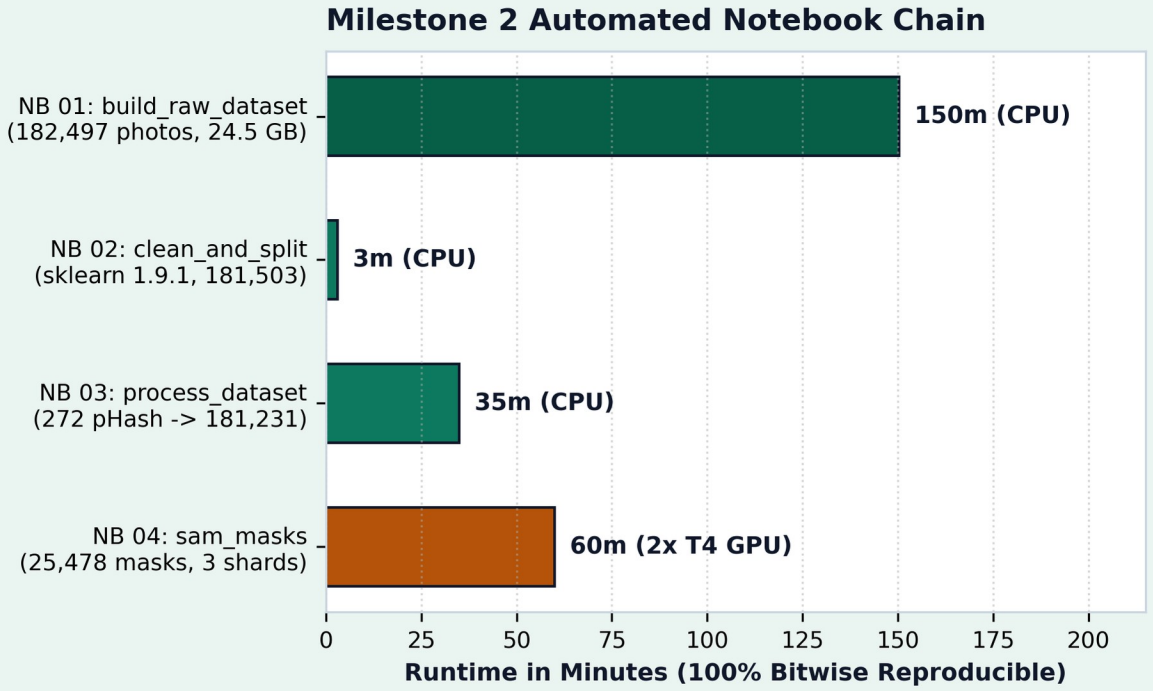
Grounding DINO + SAM 2.1 Masks

4 Iterations of Mask Refinement

- **Open SAM 2.1 Pipeline:** Uses Grounding DINO ('a snake.') + SAM 2.1 on training photos only (0.39s/photo on 2x T4).
- **Hand & Glove Subtraction:** Keeps single best snake box and subtracts hands/gloves detected in a separate pass.
- **486/500 Pilot Success:** Found snakes in 486/500 pilot photos (95 flagged); scaling to 25,478 photos across 3 shards.







7. HOSTED DATASETS & NOTEBOOK PIPELINE (SECTION 7 & 8)

100% Reproducible Kaggle Chain

End-to-End Automated Rebuild

- **3 Private Kaggle Datasets:** puneetbajajiitm/serpentguard-indian-snakes, m2-splits, and indian-snakes-processed.
- **Notebooks 01 to 04 Chain:** 01 Raw (2.5h CPU) → 02 Split (3m) → 03 Process (35m) → 04 SAM Masks (1h/shard on 2x T4).
- **Exact Bitwise Match:** Pinning scikit-learn 1.9.1 (seed 42) reproduced 100% of 181,503 split rows on 29 Oct 2026.

8. PRODUCT UI MOCKUPS & IMMEDIATE NEXT STEPS (DUE 5 NOV)

SerpentGuard AI Interface & Milestone 3 Roadmap

Offline Android UI (25–30 MB Target)

- **Confident In-Scope Match:** Top-3 species + genus + WHO Category 1/2 badge + visual cues.
- **Explicit 'Unknown' State:** 81 unknown-pool species & blurry views trigger safe abstention.
- **Fixed WHO Guidance & SOS:** Quotes WHO SEARO 2016 verbatim + one-tap emergency Bite SOS.

Milestone 3 Plan (Due 5 Nov 2026)

- **Audit 25,478 SAM Masks:** Hand-check flagged masks (review.csv) & 200-photo label noise sample.
- **Multi-Task Student Model:** Train shared encoder with mask, species (Tier-1/2), genus & WHO heads.
- **Android Feasibility Study:** Benchmark accuracy, mask quality, int8 size & latency on fixed subset.

181,231 Verified Photos Ready for Multi-Task Edge Training.

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